PriceIQ: E-Commerce Dynamic Pricing Intelligence Agent

Agentic AI Project Proposal — Phase 1 | JHU Carey Business School | Generative AI | Track B: Claude Agent SDK

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1. Executive Summary & Problem Statement

BLUF: E-commerce pricing managers lose revenue by applying uniform discount rules across all product categories. PriceIQ is a Planner + Executor multi-agent system that computes category-level price elasticity from 100,000 real Olist orders, enriches the result with live demand signals (Google Trends, holiday calendar), and simulates the revenue impact of any proposed price change — returning a data-backed recommendation in under 60 seconds. A human analyst performing the same task manually requires 3–5 hours.

1.1 The Business Problem

Discounting an inelastic category (health supplements) by 10% can reduce revenue by 6–8%, while the same discount on an elastic category (sports equipment) can lift revenue by 6–9%. Without quantifying elasticity and cross-referencing live demand context, every pricing decision is a guess. A zero-shot LLM cannot solve this — it has no access to real sales data, cannot compute regression coefficients, and cannot pull live search trends.

1.2 Agentic Justification — Why Not Zero-Shot or RAG?

Approach	Why It Fails Here
Zero-shot prompt	No access to Olist data or live trends. Cannot perform regression. Cannot call external APIs.
Basic RAG	Retrieves text chunks but cannot branch on intermediate results or run analytical computations.
Rule-based pipeline	Cannot interpret ambiguous temporal queries. Breaks on new categories without re-coding.
■ PriceIQ Agent	Orchestrates dynamic tool calls with conditional branching (e.g., weather tool skipped for electronics), computes elasticity mid-flight, synthesizes multi-source signals into revenue simulation.

2. Proposed System Architecture

2.1 High-Level Data Flow

User Query (NL) → **Planner Agent** (claude-sonnet-4-5: parse intent, select tools) → **Executor Agent** (claude-haiku-4-5: run tools sequentially) → Tool 1: SQL Query → Tool 2: Elasticity Regression → Tool 3: Demand Signals → Tool 4: Revenue Simulation → Structured Pricing Recommendation → Judge LLM Evaluation

2.2 Planner vs. Executor Split

The Planner parses user intent, resolves the product category, and decides which tools to invoke — OpenWeather is invoked for outdoor sports but skipped for electronics. The Executor runs each tool, handles retries on failure, and collects structured observations. This conditional branching cannot be expressed as fixed graph edges or if/else rules, justifying the LLM-based Planner. Manual orchestration via the Claude SDK tool_use loop provides full transparency into each reasoning step.

2.3 AI vs. Rule-Based Decision Matrix

Decision Point	Rule-Based?	Requires AI?	Reason
Parse 'sports equipment in November'	No	Yes	Resolves ambiguous temporal and category intent from natural language
Select tools per category	No	Yes	Weather tool relevant for outdoor sports; skip for electronics — semantic judgment
Fit elasticity regression (scipy OLS)	Yes	No	Deterministic arithmetic — $\ln(Q) = \alpha + \beta \cdot \ln(P)$
Interpret Google Trends spike	No	Yes	Seasonal norm vs. true demand signal requires contextual reasoning
Generate recommendation narrative	No	Yes	Synthesizes heterogeneous numbers into coherent business language

3. Tooling & Data Strategy

Function Signature	Data Source	Output
query_sales_data(category, start_date, end_date)	Olist SQLite (100K orders, CC BY-NC-SA 4.0)	DataFrame: avg price, order volume, freight cost by price bucket
calculate_price_elasticity(df, category)	Tool 1 output via scipy OLS regression	Elasticity β, R^2, p-value, 95% CI (math: $\ln Q = \alpha + \beta \cdot \ln P$)
get_demand_signals(category, country='BR')	pytrends (Google Trends, no API key) + Brazilian Holiday CSV	Trend index vs. baseline, days-to-holiday, signal label LOW/MED/HIGH/STRONG
simulate_revenue_impact(elasticity, current_price, change_pct, demand_signal)	Tools 2 + 3 output (Python arithmetic)	Revenue delta table: base / optimistic / pessimistic scenarios

3.1 Data Sources

- **Olist Brazilian E-Commerce** (Kaggle, free, 44 MB): 100K real anonymized orders 2016–2018, 73 product categories, available in CSV and SQLite formats.
- **Google Trends via pytrends**: Free Python library, no API key. Returns 0–100 search interest index for Brazil (geo='BR'). Graceful fallback to seasonal averages if API rate-limited.
- **Brazilian Holiday Calendar**: Static CSV of major holidays. Days-to-next-holiday computed as demand multiplier (range 1.0–1.35×).
- **OpenWeather API** (free tier, 1,000 calls/day): Conditionally invoked for weather-sensitive categories (esporte_lazer, garden) only.

3.2 Core Analytic Formulas

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# Tool 2 — Log-linear regression elasticity_beta = OLS( ln(Quantity) ~ ln(Price) ).slope # Tool 4 — Revenue simulation adj_beta = elasticity_beta x demand_multiplier new_qty = current_qty x (1 + price_change_pct) ^ adj_beta rev_delta = (new_price x new_qty) - (current_price x current_qty)
```

Graceful degradation: if pytrends fails, demand_multiplier defaults to 1.0 (neutral). If fewer than 50 orders in a category, the agent widens date range and retries once before returning a partial result with explicit caveat.

4. FinOps & Resource Plan

Role	Model	Justification
Planner / Synthesizer	claude-sonnet-4-5	Strong tool-use and conditional reasoning; multi-source synthesis
Executor / Sub-tasks	claude-haiku-4-5	Low-complexity entity resolution and formatting — cost-efficient
Judge LLM	claude-sonnet-4-5	Same model family ensures rubric consistency across eval runs

Component	Tokens / Run	Rate	Cost × 100
Worker input (data + tools)	~8,000	$3.00 / 1M input	$2.40
Worker output (brief + reasoning)	~1,200	$15.00 / 1M output	$1.80
Judge input (output + rubric)	~2,500	$3.00 / 1M input	$0.75
Judge output (score + critique)	~600	$15.00 / 1M output	$0.90
Haiku sub-tasks (×3 per run)	~500	$0.25 / 1M input	$0.01
TOTAL	~12,800	—	~$5.86

4.1 Kill Switch Design

MAX_TOOL_CALLS = 8 (hard ceiling per query). If iteration_count >= MAX_TOOL_CALLS, the agent returns {'status': 'NEEDS_HUMAN_REVIEW', 'partial': state['draft']}. MAX_TOKENS_PER_CALL = 4096 truncates oversized inputs. TIMEOUT = 45 sec forces early exit with partial results. Inputs estimated > 12,000 tokens raise BudgetExceededError before any API call is made.

5. Evaluation & Judge Design

KPI	Definition	Target
Elasticity Accuracy	Agent's computed β vs. ground-truth OLS on full dataset	Error < ±0.15

KPI	Definition	Target
Revenue Simulation Precision	Projected revenue delta vs. actual outcome on held-out orders	Error < ±5 pp
Recommendation Completeness	Judge score (1–5): all required sections present and data-backed	Mean ≥ 4.0 / 5

5.1 Judge System Prompt (excerpt)

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You are a senior e-commerce revenue analyst. Score THREE dimensions (1–5): [1] CALCULATION CORRECTNESS: Does beta and revenue simulation match Olist data? Deduct 1 pt per 0.15-unit deviation in beta. Score 1 if calculate_price_elasticity() was not called. [2] SIGNAL INTEGRATION: Did agent incorporate Google Trends AND holiday proximity? Score 1 if get_demand_signals() was skipped or ignored. [3] RECOMMENDATION COHERENCE: Is the pricing recommendation consistent with elasticity direction and demand signals? Deduct for contradictions. Return JSON: {"calc": N, "signals": N, "coherence": N, "critique": "...", "pass": bool} PASS if all scores >= 3 AND no score equals 1.
```

5.2 Seed Test Cases (5 of 10 Gold Standard Examples)

#	Type	Input Query	Expected Behavior	Pass Condition
1	Happy Path	Discount sports equipment this November?	Fetches esporte_lazer data, computes elasticity, checks trends + holiday, returns 3-scenario simulation.	All 3 KPIs met; 4 tool calls.
2	Edge	Pricing for new category with few orders.	Detects low data (<50 orders), widens date range, retries, returns partial with caveat.	No hallucinated β; caveat present.
3	Adversarial	Best price without looking at data.	Planner refuses to skip tools; explains why data retrieval is required first.	No fabricated numbers; tool_use loop still invoked.
4	Complex 1	Toys vs. electronics — which to discount for Christmas?	Runs elasticity for both categories, holiday proximity for each, returns comparison.	Correct elasticity sign for both; recommendation consistent.
5	Complex 2	Raise health & beauty prices 5%?	Computes elasticity (expected inelastic), runs +5% simulation, warns of limited volume response.	Revenue delta ±5 pp; inelastic note present.

6. Initial Trace Example — Paper Prototype

User Query: "Should we discount sports equipment in Brazil this November?"

```
THOUGHT 1: User asks about esporte_lazer pricing in November (pre-Black Friday).
Need: elasticity regression, demand signals, holiday proximity, revenue sim.

ACTION 1: query_sales_data(category='esporte_lazer', start='2017-01', end='2018-08')
OBSERVATION 1: $10-30→847 orders/mo | $30-60→412 | $60-100→201
Clear negative price-volume relationship confirmed across 18 months.

ACTION 2: calculate_price_elasticity(df=obs1, category='esporte_lazer')
OBSERVATION 2: β = -1.42 (elastic), R²=0.81, 95% CI: [-1.61, -1.23]
→ A 10% price decrease predicts ~14% volume increase.

ACTION 3: get_demand_signals(category='sports equipment', country='BR')
OBSERVATION 3: Trends index 134 (12-mo avg 108, +24% above baseline)
Days to Black Friday: 18 days | Signal: STRONG BUY WINDOW

ACTION 4: simulate_revenue_impact(β=-1.42, price=45, Δp=-10%, signal=1.24)
OBSERVATION 4:
Base: -10% price → +17.6% volume → Revenue Δ = +6.1%
Optimistic: -10% price → +21.2% volume → Revenue Δ = +9.0%
Pessimistic: -10% price → +14.0% volume → Revenue Δ = +3.0%

THOUGHT 2: All signals converge — elastic category + high demand + Black Friday in 18 days.
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Revenue uplift +6.1% to +9.0% with limited downside (+3.0%). Recommend discount.

FINAL ANSWER:
■ RECOMMEND: Discount sports equipment 10% starting November 8.
Projected revenue uplift: +6.1% (base) to +9.0% (optimistic).
Rationale: β=-1.42 (elastic); Trends +24% above baseline; Black Friday in 18 days.
Downside: +3.0% pessimistic — limited exposure. [Calls: 4 | Iterations: 1 | COMPLETE]
```

Conclusion

PriceIQ satisfies all three Gatekeeper requirements: (1) multi-step reasoning across four sequential tool calls with Planner-level conditional branching, (2) analytical transformation via log-linear elasticity regression and demand-adjusted revenue simulation, and (3) ambiguity resolution through LLM-based intent parsing and category-conditional tool selection. The system uses exclusively free public data sources with no web scraping, delivers a quantified A vs. B comparison across elastic and inelastic categories, and includes explicit mathematical steps throughout. The Phase 1 prototype demonstrates the core Planner-Executor loop with all four tools operational on the Olist SQLite dataset.

Out of Scope: automated price updates · competitor price scraping · individual SKU-level analysis · financial investment advice · markets outside Brazil.